

PEST CONTROL

Agricultural Science - Comprehensive Lesson Guide

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Definition of a Pest: A pest is any living organism that causes harm to humans, crops cultivated by humans, farm animals, or materials/property utilized by humans.

1. Categorization of Agricultural Pests

Crop pests can be categorized into four main groups based on their characteristics and the nature of the damage they cause:

- **Weeds:** Unwanted plants growing in crop fields (e.g., Touch-me-not / *Mimosa pudica*, Lantana / *Lantana camara*).
- **Pathogenic Organisms:** Microorganisms causing diseases (e.g., Fungi, Bacteria, Viruses, Phytoplasmas).
- **Insects:** Insect pests that damage plant parts (e.g., Paddy bug, Fruit fly).
- **Non-Insect Animals:** Other animal pests (e.g., Snails, Mites, Birds, Mammals).

2. Weeds in Agriculture

Any plant growing in a cultivated field other than the intended crop is referred to as a **weed**.

Benefits of Weeds

- Helps reduce soil erosion.
- Utilized for compost production.
- Used as medicinal plants (e.g., Nut grass / *Cyperus rotundus*, Goosegrass / *Eleusine indica*).
- Serves as animal feed/fodder (e.g., specific grasses).
- Used as human food/leafy vegetables (e.g., *Kirihenda*, *Girapala*).
- Used in landscaping and ornamental ground cover.

General Characteristics of Weeds

- Ability to grow vigorously even under adverse environmental conditions.
- Rapid propagation and spread.
- Ability to survive during harsh/unfavorable periods.
- High resistance to pests and diseases.
- Short life cycle.
- Possess strong and extensive root systems.
- Dispersed through various effective mechanisms.

Harmful Effects of Weeds

- Competes with crops, weakening them and reducing total yield.
- Reduces the quality of the harvested crop yield (e.g., weed seeds like *Bajiri* or *Maruk* mixing with harvested paddy).
- Makes field agricultural operations difficult (e.g., thorny weeds like Touch-me-not).
- Acts as alternative host plants for various pathogens.
- Blocks irrigation pathways and water channels (aquatic weeds).
- Helps insect pests complete their life cycle by providing shelter/food (e.g., Paddy bug).

Invasive Weeds

Foreign (alien) plants that spread very rapidly and pose a severe threat to the existence of native plant communities in the ecosystem.

Examples: Guinea grass, Wedelia, Giant Mimosa (*Yodha Nidikumba*), Lantana (*Gandapana*).

3. Classification of Weeds

A. By Morphology (Structure)

Broad-leaved Weeds	Grasses (Gramineae)	Sedges (Cyperaceae)
• Leaves are broad with reticulate (netted) venation. • Stems are generally round, sometimes square, and solid inside. • Flowers occur singly or in inflorescences at terminal or lateral branches. <i>Examples: Monarakudumbiya, Kuppameniya, Water hyacinth (Japan Jabara).</i>	• Leaves are narrow, long with parallel venation. • Stems are cylindrical and hollow inside. • Inflorescence forms at the stem apex. <i>Examples: Batadella, Balathana, Iluk, Mana.</i>	• Leaves are narrow, long with parallel venation. • Stems are triangular in cross-section and solid (no cavity). • Inflorescence forms at the apex with 3 leaves extending in 3 directions. <i>Examples: Kalanduru, Kudametta, Thunhiria.</i>

B. By Habitat (Growth Location)

- **Terrestrial Weeds:** Grow on land (e.g., *Monarakudumbiya, Kuppameniya, Godamaruk, Touch-me-not, Thuttiri*).
- **Aquatic Weeds:** Grow in water (e.g., *Water hyacinth / Japan Jabara, Diya Siyambala, Diya Habarala, Kudametta*).

C. By Life Span

- **Annual Weeds:** Complete their entire life cycle and die within one year or less (e.g., *Kuppameniya, Monarakudumbiya*).
- **Perennial Weeds:** Take more than one year to complete their life cycle and survive across seasons (e.g., *Attadi, Atawara, Kalanduru, Lantana*).

4. Weed Management Strategies

Strategy 1: Keeping existing weeds below the Economic Threshold Level (ETL)

- Selecting appropriate cropping systems or crop rotations.
- Proper primary and secondary land preparation.
- Maintaining proper spacing during planting and staking.
- Manual weeding (hand weeding) and hoeing.
- Applying soil mulch.
- Removing aerial parts of weeds.
- Growing cover crops.
- Impounding water in paddy fields (flooding).
- Using mechanical weeder/machines.
- Intercropping/mixed cropping.

Strategy 2: Preventing the introduction of new weeds into the field

- Planting clean, high-quality seeds free from weed seeds.
- Using green manure or compost fertilizer that is free from weed seeds.
- Keeping irrigation canals and water inlets clean and weed-free.
- Using clean farm machinery and implements.
- Maintaining weed-free surroundings around the farm field.
- Enforcing plant quarantine acts and regulations.

5. Classification of Chemical Herbicides

• Based on Time of Application:

- *Pre-emergence Herbicides*: Applied before weed seeds germinate/emerge. Applied to soil during land preparation.
- *Post-emergence Herbicides*: Sprayed onto foliage after weeds have emerged from the soil.

• Based on Mode of Action:

- *Contact Herbicides*: Kills only the plant tissues directly coming into contact with the chemical.
- *Systemic (Translocated) Herbicides*: Absorbed through roots/leaves and translocated throughout the entire plant to kill it completely.

• Based on Selectivity:

- *Non-selective Herbicides*: Destroys all types of vegetation (broad-leaved, sedges, and grasses).
- *Selective Herbicides*: Controls specific weed types while leaving crops undamaged (e.g., controlling broad-leaved weeds in grass crops).

6. Plant Diseases

A plant disease is an abnormal physiological process caused by biotic or abiotic factors that disrupts the plant's normal growth, development, or functions.

Causative Agents of Plant Diseases

1. **Fungal Diseases:** Spread via spores carried by wind, water, soil/planting media, animals, and farm tools.
Symptoms: Leaf spots, blights, wilting, galls, wet/dry rots, rusts, powdery mildew.
2. **Bacterial Diseases:** Spread via water, air, farm tools, and animals.
Symptoms: Water-soaked spots, bacterial blight, bacterial wilt, soft rot, galls/knots, scabs.
3. **Viral Diseases:** Transmitted via insect vectors (aphids, whiteflies) and mechanical injury.
Symptoms: Leaf yellowing (chlorosis), stunting, vein clearing, leaf/flower distortion, mosaic patterns.
4. **Nematode Diseases:** Microscopic roundworms attacking roots, leaves, flowers, and fruits.
Symptoms: Root knots/galls, plant stunting/weakening, root branching/shortening, leaf yellowing.
5. **Phytoplasma Diseases:** Parasitic organisms living in phloem tissues.
Symptoms: Virescence (greenish flowers), yellowing leaves, stunting, rosette formation (phyllody/witches' broom).

7. Plant Disease Control Triangle & Strategies

Disease development requires three factors: a **susceptible host plant**, a **virulent pathogen**, and a **favorable environment**.

Control Strategies

- Reducing the pathogen population present in the field.
- Inhibiting pathogen growth and development on the host plant.
- Creating environmental conditions unfavorable for the pathogen.
- Preventing entry of new pathogens from outside into the field.

Major Plant Diseases & Management

Fungal / Oomycete

1. Damping Off

Pathogens: *Pythium*, *Rhizoctonia*

Symptoms: Affects germinating seeds and nursery seedlings; poor seed germination; shriveled cotyledons with black spots; seedling collar/stem rots and collapses near soil level.

Control: Plant healthy seeds; seed treatment; soil sterilization; increase seedling spacing in nurseries; apply fungicides.

Fungal

2. Anthracnose

Pathogen: *Colletotrichum*

Symptoms: Irregular brown or black spots on leaves; spots coalesce; water-soaked sunken lesions on fruits.

Control: Cultivate resistant varieties; prune and burn infected twigs/leaves; avoid wetting foliage during irrigation; apply fungicides.

Bacterial

3. Bacterial Wilt (Hitu Mahrma)

Crops Affected: Solanaceae (Tomato, Eggplant, Chili) and Cucurbitaceae.

Symptoms: Rapid wilting and death of plant; sticky bacterial ooze visible when cut stem is immersed in water; adventitious roots emerge from stem.

Control: Crop rotation; growing resistant varieties; ensuring proper soil drainage; rogueing/removing infected plants.

Viral

4. Mosaic Virus

Crops Affected: Okra, Chili, Papaya, Tomato, Tobacco.

Symptoms: Mottled light green and dark green mosaic patterns on leaves; plant stunting; malformed flowers and fruits.

Control: Crop rotation; plant resistant varieties; remove host weeds; control insect vectors.

Nematode

5. Root-Knot Disease

Crops Affected: Vegetables, ornamental plants.

Symptoms: Swelling/galls on root system; weak plant growth; wilting during hot sunshine.

Control: Soil solarization/treatment; crop rotation with non-hosts; planting resistant crop varieties.

8. Insect and Animal Pests

Insect Pest Metamorphosis

- **Complete Metamorphosis (Holometabolous):** Egg → Larva → Pupa → Adult (e.g., Butterfly, Moth, Beetle).
- **Incomplete Metamorphosis (Hemimetabolous):** Egg → Nymph → Adult (e.g., Bugs, Aphids).

Modes of Damage by Insect Pests

- Chewing and biting plant tissues.
- Piercing and sucking plant sap.
- Rolling/folding leaves to create nests.

- Acting as vectors for viral and phytoplasma diseases.

Non-Insect Animal Pests

- **Mites:** Microscopic; suck cell sap from leaves causing curling/browning.
- **Birds:** Parrots, Barbets damaging ripe fruits and grains.
- **Molluscs:** Snails and slugs eating tender shoots and foliage.
- **Mammals:** Porcupines, Bats, Rabbits damaging underground tubers, fruits, and stems.

9. Key Agricultural Pests & Specific Control

Pest Name	Damaging Stage	Nature of Damage	Control Methods
Brown Plant Hopper (BPH)	Nymph & Adult	Sucks sap from base of paddy plants causing "hopper burn".	Synchronized planting; resistant varieties (Bg352, Bg397); chemical sprays if needed.
Fruit Fly	Larva (Maggot)	Female pierces fruit skin to lay eggs; maggots feed on flesh causing fruit rot.	Deep burial of fallen fruits; pheromone traps; fruit bagging; insecticides.
Cucurbit Fly (II Massaa)	Larva (Maggot)	Attacks cucurbit fruits; maggots burrow inside flesh.	Field sanitation; fruit wrapping; protein bait traps; chemical control.
Aulacophora Beetle	Larva & Adult	Feeds on cucurbit leaves and roots; lays eggs near stem base.	Hand/net collection of adults; soil treatment; insecticides.
Epilachna Beetle	Larva & Adult	Scrapes leaf tissues leaving a skeletonized net-like leaf pattern.	Hand picking; destroying egg masses; applying recommended insecticides.
Red Palm Weevil	Larva (Grub)	Grubs bore into coconut trunk tissues, causing leaf wilting and palm death.	Tar application on stem wounds; frequent inspection of palms under 12 yrs; pheromone traps.

10. Integrated Pest Management (IPM) & Pesticides

Integrated Pest Management (IPM): An eco-friendly approach combining cultural, mechanical, biological, and quarantine methods to manage pest populations below economic injury levels, keeping chemical control as a last resort.

Classification of Pesticides

- **By Target Organism:** Herbicides, Fungicides, Insecticides, Acaricides (Miticides), Molluscicides, Nematicides.
- **By Formulation:** Aqueous solutions, Emulsifiable concentrates (EC), Wettable powders (WP), Soluble powders (SP), Granules (G).

Safe Pesticide Usage Guidelines

- Transport chemicals carefully in sealed original containers.
- Select the exact target pesticide recommended for the pest.
- Mix strictly at the recommended dosage and concentration.
- Spray at prescribed time intervals wearing protective gear.
- Safe disposal: Bury empty chemical containers safely.
- Bathe thoroughly after application.
- Observe Pre-Harvest Intervals (PHI) strictly before picking produce.

Environmental & Health Issues of Chemical Pesticides

- Contamination of ground and surface water systems.
- Destruction of beneficial natural enemies (e.g., spiders, ladybird beetles, dragonflies, pollinators).
- Pesticide residue accumulation entering human food chains, causing chronic illnesses (e.g., cancers, kidney diseases).

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